Circuit Variables

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Structure

- Electrical Engineering: An Overview
- The International System of Units
- Circuit Analysis: An Overview
- Voltage and Current
- The Ideal Basic Circuit Element
- Power and Energy

Electrical Engineering: An Overview



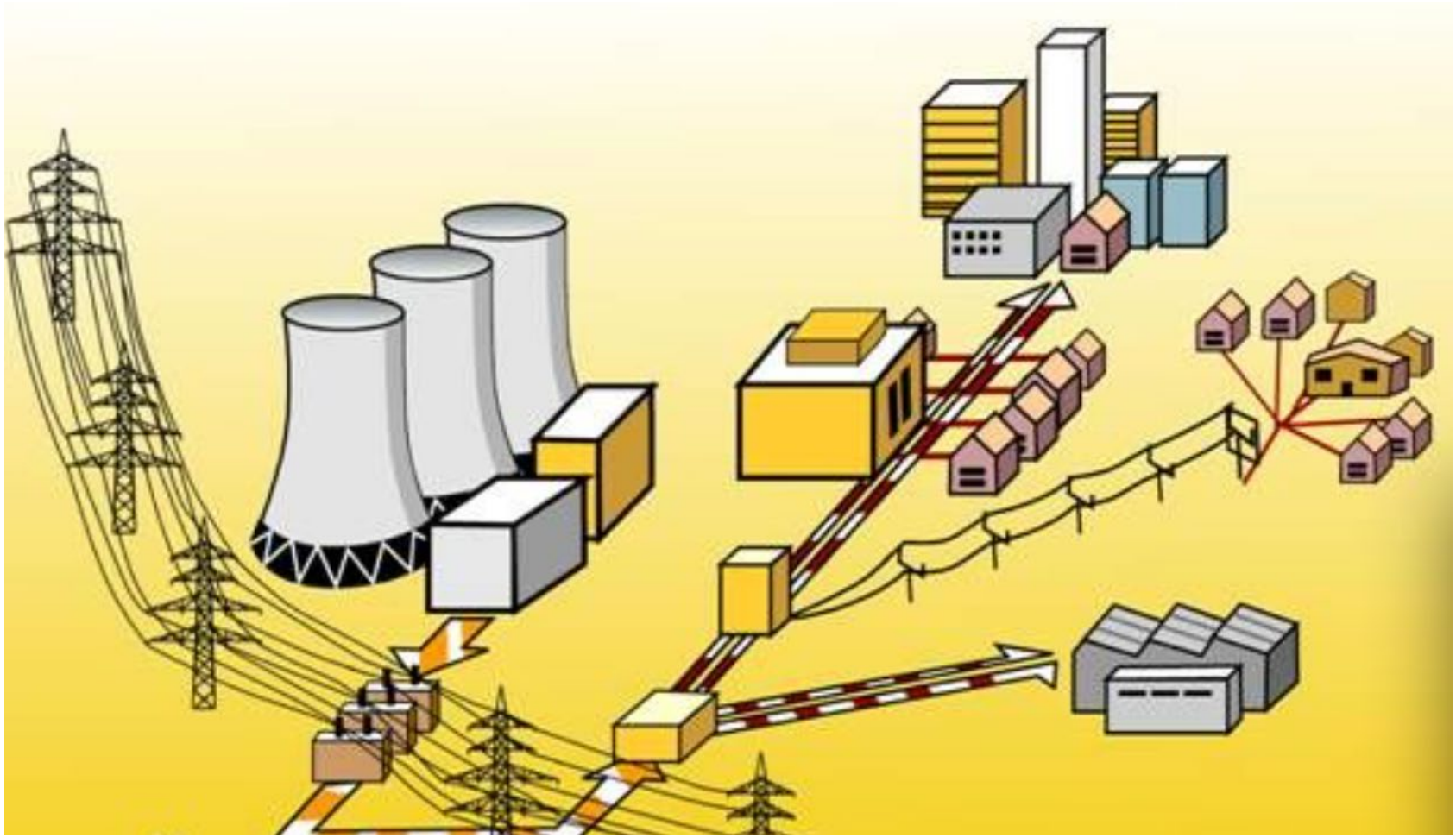








Power Systems



Anything in Common?

- An **electric circuit** is a *mathematical model* that approximates the behavior of an actual electrical system.

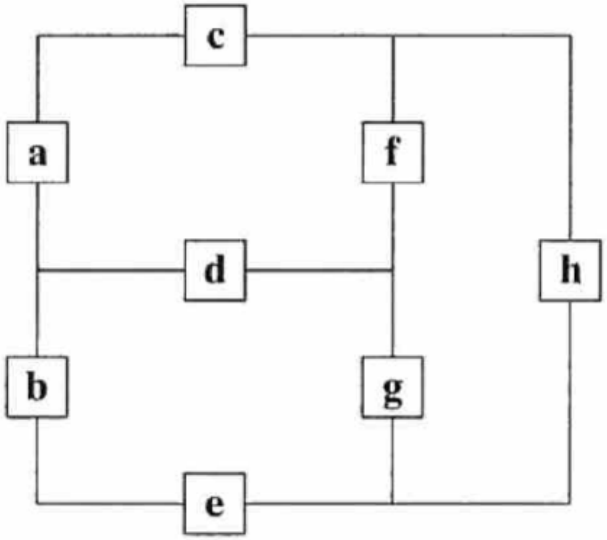
- Here, we use circuit theory, rather than electromagnetic field theory, to study a physical system represented by an electric circuit, based on the following three assumptions:
  - Electrical effects happen instantaneously throughout a system (***lumped-parameter system***)
  - The net charge on every component in the system is always zero
  - There is no magnetic coupling between the components in a system

Problem Solving

- Identify what's given and what's to be found.
- Sketch a circuit diagram or other visual model.
- Think of several solution methods and decide on a way of choosing among them.
- Calculate a solution.
- Use your creativity.
- Test your solution.

Balancing Power





- **a** and **b**: electrical source to the home;
- **c**, **d**, and **e**: the wires that carry the electrical current from the source to the devices in the home requiring electrical power;
- **f**, **g**, and **h**: lamps, televisions, hair dryers, refrigerators, and other devices that require power.

The International System of Units

Quantity	*Basic unit*	*Symbol*
Length	meter	m
Mass	kilogram	Kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Derived Units in SI

Quantity	Unit Name (Symbol)	Formula
Frequency	hertz (Hz)	s^{-1}
Force	newton (N)	$kg \cdot m/s^2$
Energy or work	joule (J)	$N \cdot m$
Power	watt (W)	J/s
Electric charge	coulomb (C)	$A \cdot s$
Electric potential	volt (V)	J/C
Electric resistance	ohm (Ω)	V/A
Electric conductance	siemens (S)	A/V
Electric capacitance	farad (F)	C/V
Magnetic flux	weber (Wb)	$V \cdot s$
Inductance	henry (H)	Wb/A

Standardized Prefixes to Signify Powers of 10

Prefix	Symbol	Power
atto	a	10^{-18}
femto	f	10^{-15}
pico	p	10^{-12}
nano	n	10^{-9}
micro	μ	10^{-6}
milli	m	10^{-3}
centi	c	10^{-2}
deci	d	10^{-1}
deka	da	10
hecto	h	10^{2}
kilo	k	10^{3}
mega	M	10^{6}
giga	G	10^{9}
tera	T	10^{12}

Example #1

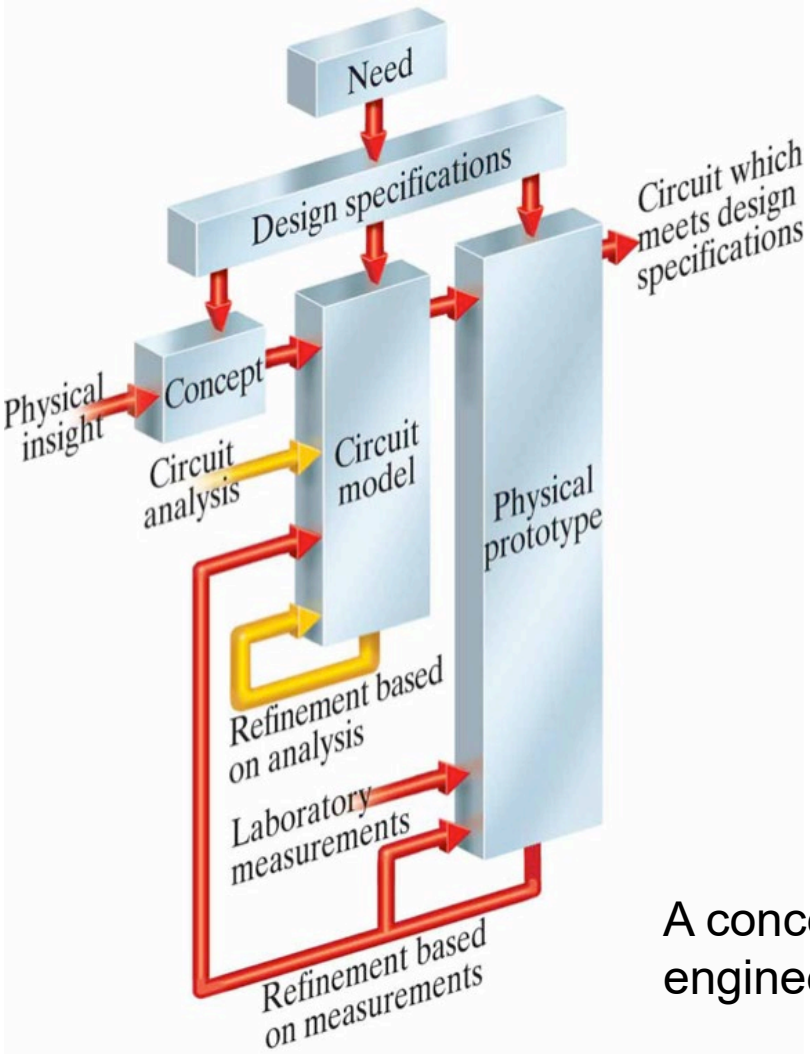
- If a signal can travel in a cable at 80% of the speed of light, what length of cable, in inches, represents 1 ns?

Solution：

$$\frac{2.4 \times 10^8 \text{ meters}}{1 \text{ second}} \cdot \frac{1 \text{ second}}{10^9 \text{ nanoseconds}} \cdot \frac{100 \text{ centimeters}}{1 \text{ meter}} \cdot \frac{1 \text{ inch}}{2.54 \text{ centimeters}}$$

$$= \frac{(2.4 \times 10^8)(100)}{(10^9)(2.54)} = 9.45 \text{ inches/nanosecond}$$

Circuit Analysis: An Overview



A conceptual model for electrical engineering design.

Voltage and Current

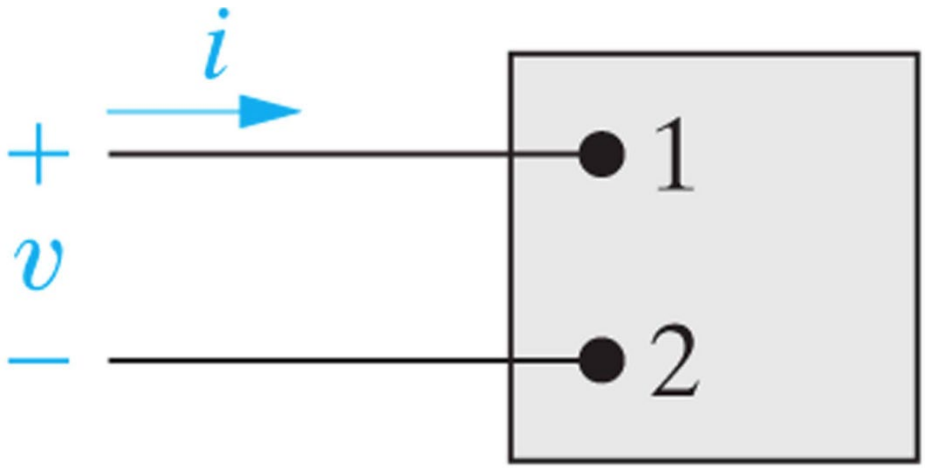
- The charge is bipolar, meaning that electrical effects are described in terms of positive and negative charges.
- The electric charge exists in discrete quantities, which are integral multiple of the electronic charge, 1.6022×10^{-19}C.
- Electrical effects are attributed to both the separation of charge (Voltage) and charges in motion (Current).

Voltage

- Whenever positive and negative *charges* (q) are separated, *energy* (w) is expended. *Voltage* (v) is the energy per unit charge created by the separation.
  - w: the energy in joules (J)
  - q: the charge in coulombs (C)
  - v: the voltage in volts (V)
  - ✓1 V is the same as 1 J/C.
- Mathematically,

$$v = dw/dq$$

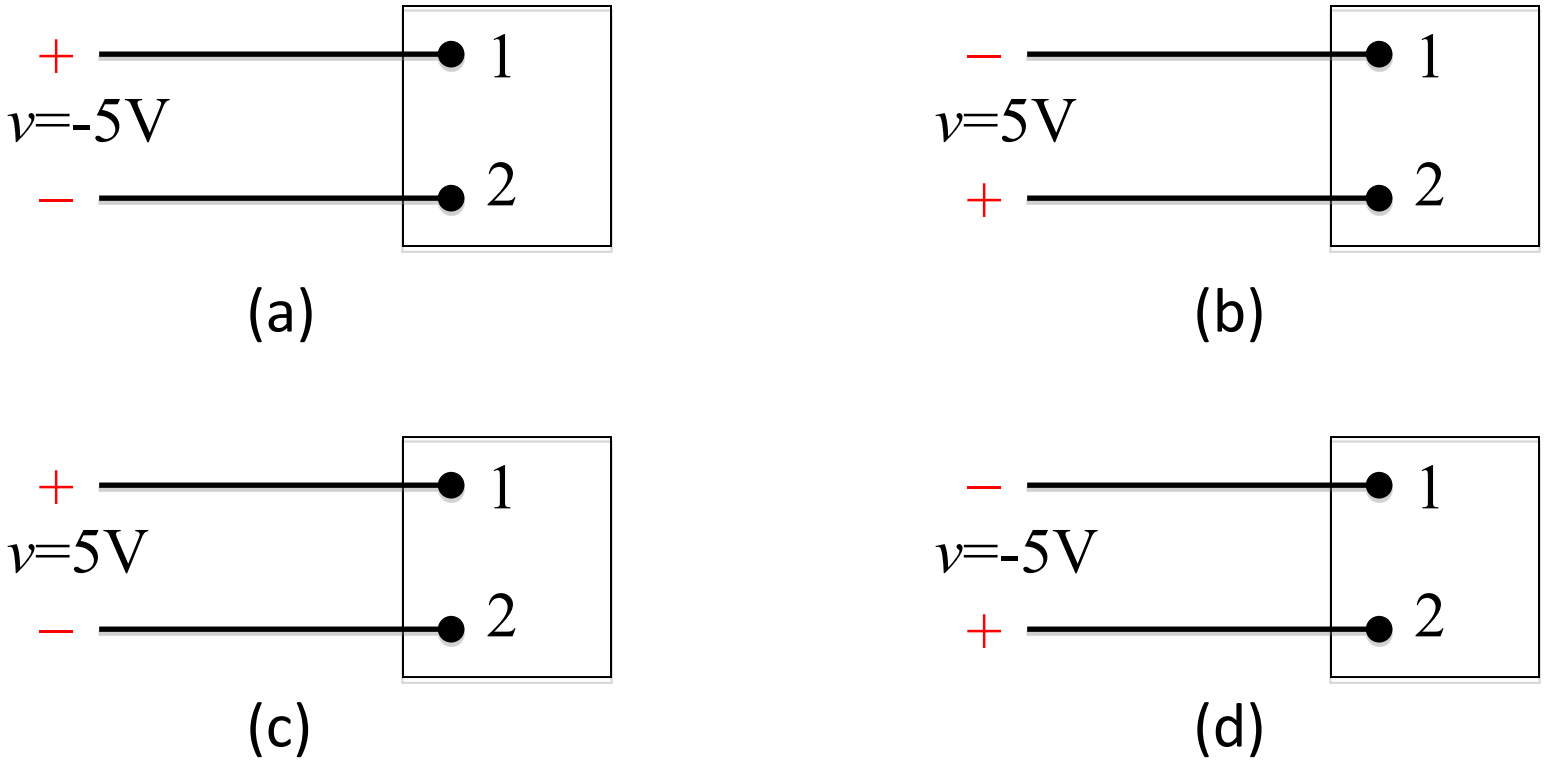
Signs of the Terminals



The placement of +sign in terminal 1 indicates that terminal 1 is v volts positive with respect to terminal 2.

Note: *A voltage can exist between a pair of electrical terminals whether a current is flowing or not.*

Example #2



(a, b) Terminal 2 is 5V positive with respect to terminal 1;
(c, d) Terminal 1 is 5V positive with respect to terminal 2.

Notes

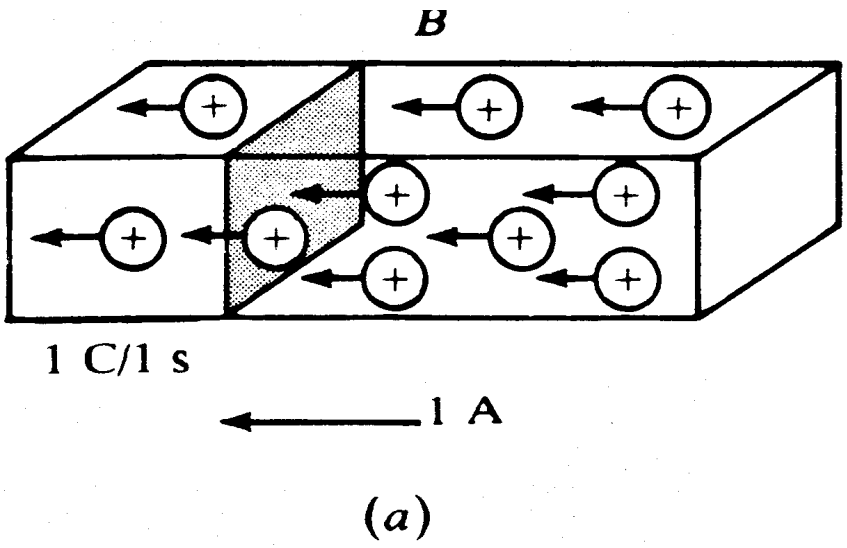
- The plus-minus pair of algebraic signs does not indicate the actual polarity of the voltage but is simple part of a convention that enable us to talk unambiguously about the voltage across the terminal pair.
- *The definition of any voltage must include plus-minus sign pair!*

Current

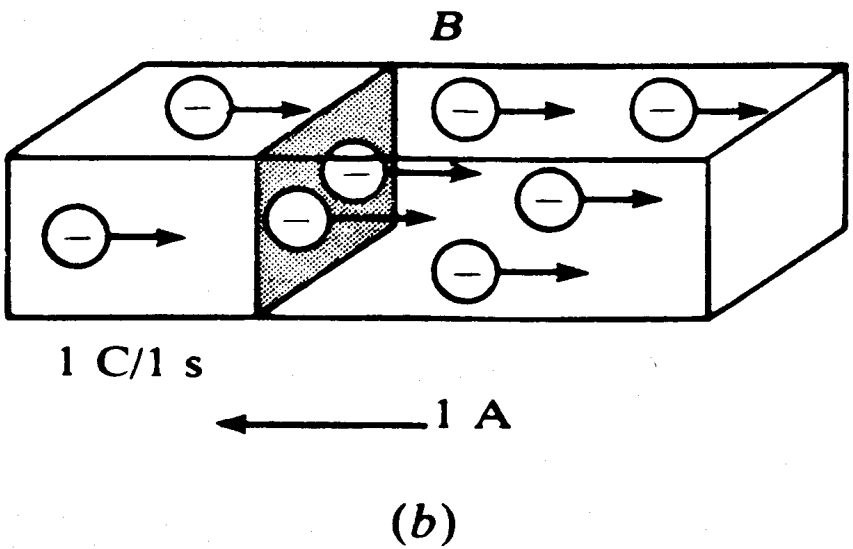
- The motion of charges forms the electric current in a wire; The current (i) has both a numerical value and a direction associated with it; It is a measure of the rate at which charge (q) is moving past a given reference point in a specified direction.
  - i: the current in amperes (A)
  - q: the charge in coulombs (C)
  - t: the time in seconds (s)
  - ✓ 1 A is the same as 1 C/s.
- Mathematically,

$$i = dq/dt$$

The Direction of Current Flow



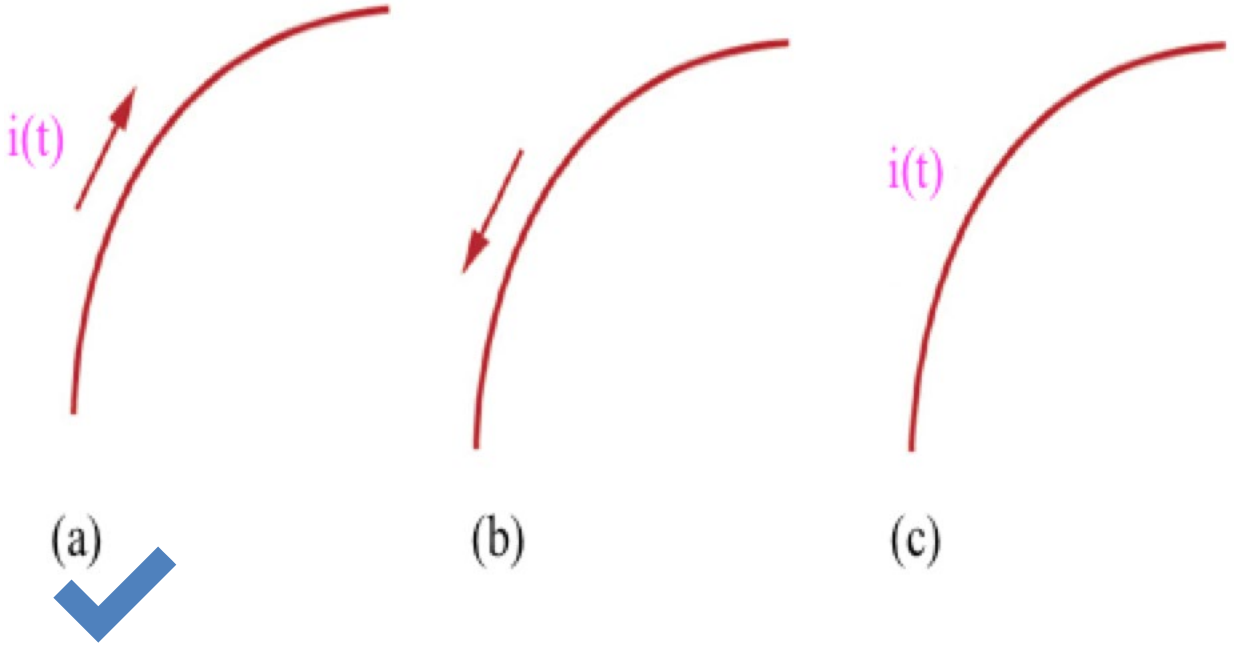
Positive ions



Negative ions

Direction is Important!

- We should pay close attention to that the arrow is a fundamental part of the definition of the current!

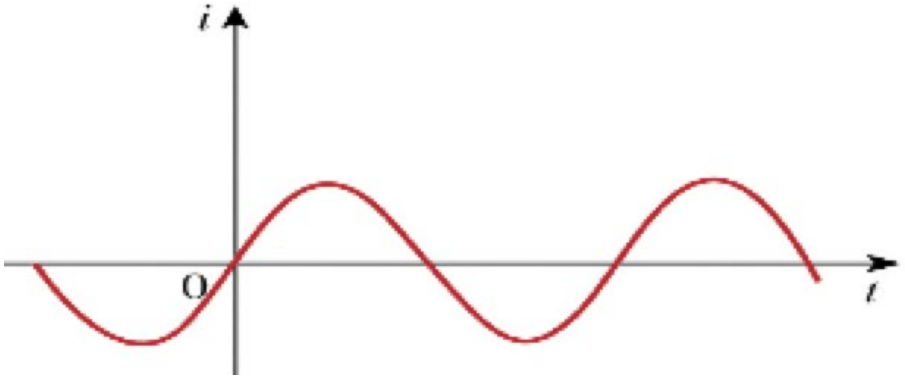


DC and AC

- A direct current (dc) is a current that remains constant with time.
- An alternating current (ac) is a current that varies sinusoidally with time. (reverse direction)

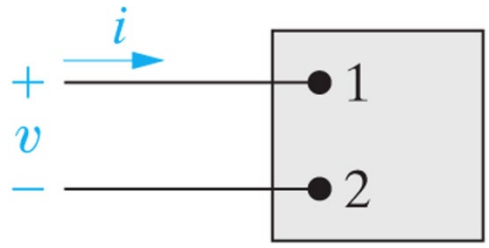


DC



AC

Example #3



No charge exists at the upper terminal of the element for $t < 0$. At $t = 0$, a 5 A current begins to flow into the upper terminal.

a) Derive the expression for the charge accumulating at the upper terminal of the element for $t > 0$.

b) If the current is stopped after 10 seconds, how much charge has accumulated at the upper terminal?

Solution:

a)From the definition of current given in Eq. 1.2, the expression for charge accumulation due to current flow is

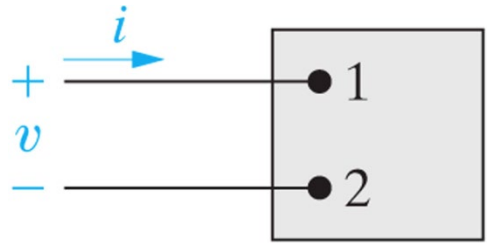
$$q(t) = \int_0^t i(x)dx$$

Therefore, we have

$$q(t) = \int_0^t 5dx = 5x\Big|_0^t = 5t - 5(0) = 5t \text{ C} \quad \text{for } t > 0$$

b)The total charge that accumulates at the upper terminal in 10 seconds due to a 5 A current is ¢(10) = 5(10) = 50 C.

Notes



Positive Value	Negative Value
v voltage drop from terminal 1 to terminal 2	voltage rise from terminal 1 to terminal 2
or	*or*
voltage rise from terminal 2 to terminal 1	voltage drop from terminal 2 to terminal 1
i positive charge flowing from terminal 1 to terminal 2	positive charge flowing from terminal 2 to terminal 1
or	*or*
negative charge flowing from terminal 2 to terminal 1	negative charge flowing from terminal 1 to terminal 2

Whenever the reference direction for the current in an element is in the direction of the reference voltage drop across the element, use a positive sign in any expression that relates the voltage to the current. Otherwise, use a negative sign.

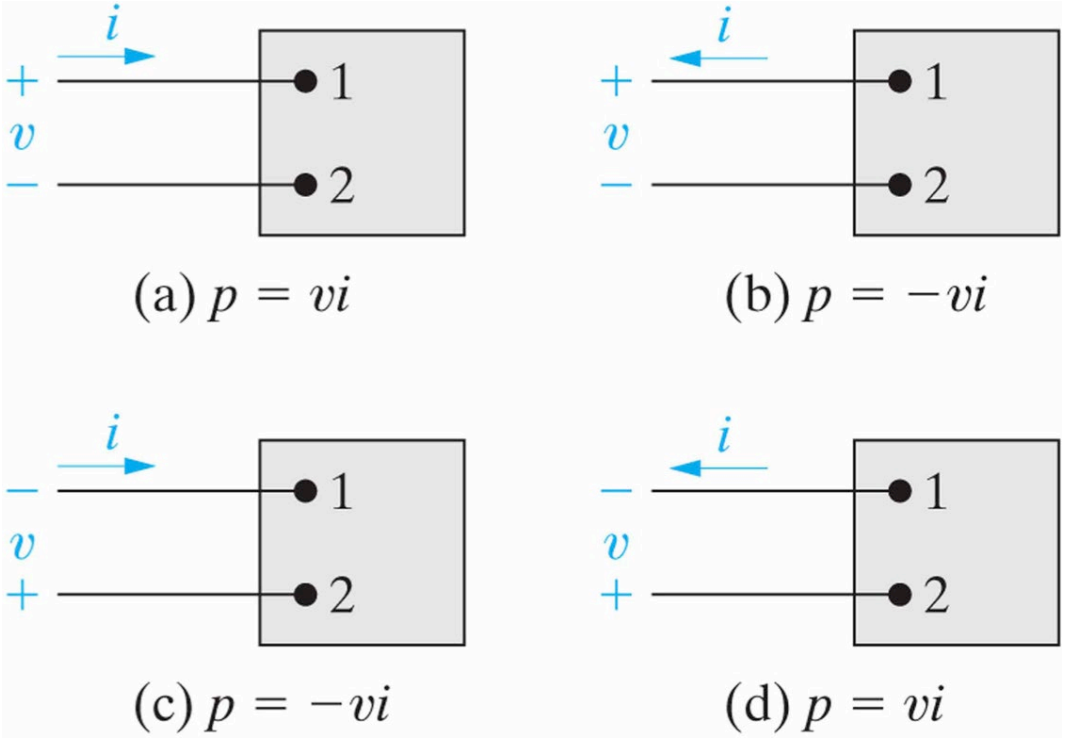
Power and Energy

- Power (p) is the time rate of expending or absorbing energy (w).
- Mathematically,

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$$

  - p: the power in watts (W)
  - w: the energy in joules (J)
  - t: the time in seconds (s)
  - q: the charge in coulombs (C)
  - v: the voltage in volts (V)
  - i: the current in amperes (A)

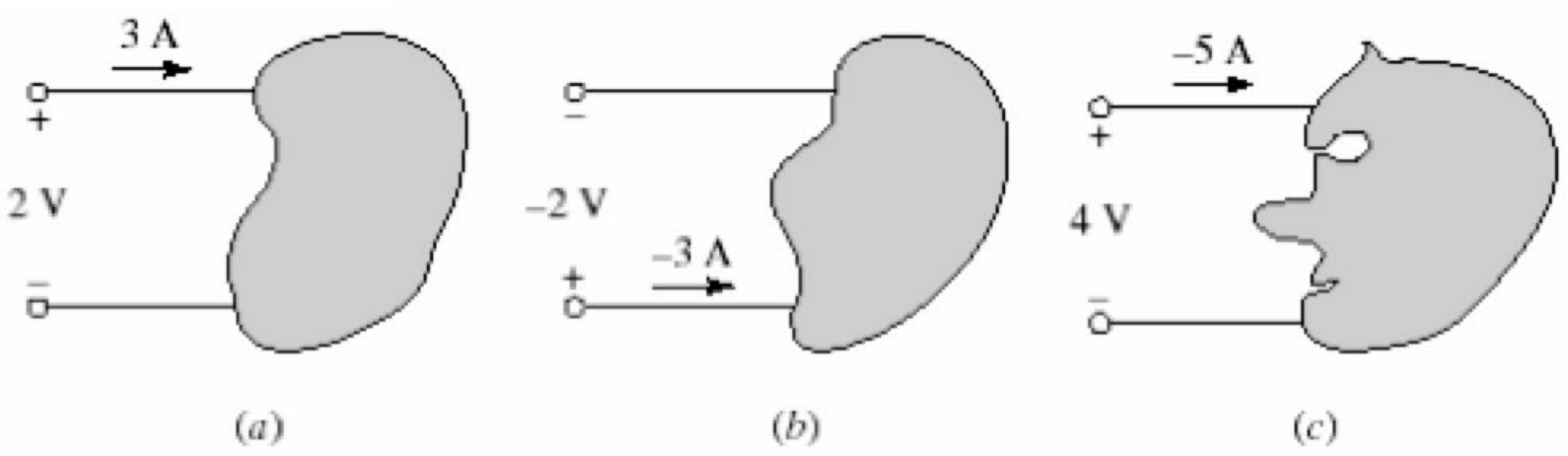
Passive Sign Convention



The algebraic sign of power is based on charge movement through voltage drops and rises. *As positive charges move through a drop in voltage, they lose energy, and as they move through a rise in voltage, they gain energy.*

Example #4

- If the power is positive (that is, if $p > 0$), power is being delivered to the circuit inside the box. If the power is negative (that is, if $p < 0$), power is being extracted from the circuit inside the box.
- Compute the power absorbed by each part in the following:



(*a*, *b*, *c*) Three examples of two-terminal elements.

Solution for the Example #4

- In (a), with +3 A flowing into the positive reference terminal, we compute

$$P = (2\ V)\ (3\ A) = 6\ W$$

of the power absorbed by the element.

- (b) shows a slightly different picture. Now we have a current of -3 A flowing into the positive reference terminal. However, the voltage as defined is negative. This gives us an absorbed power

$$P = (-2\ V)\ (-3\ A) = 6\ W$$

- In (c), we again apply the passive sign convention rulers and compute an absorbed power

$$P = (4\ V)(-5\ A) = -20\ W$$

Since we computed a negative absorbed power, this tells us that the element in the Figure is actually supplying +20 W (i.e., it's a source of energy).

Balancing Power

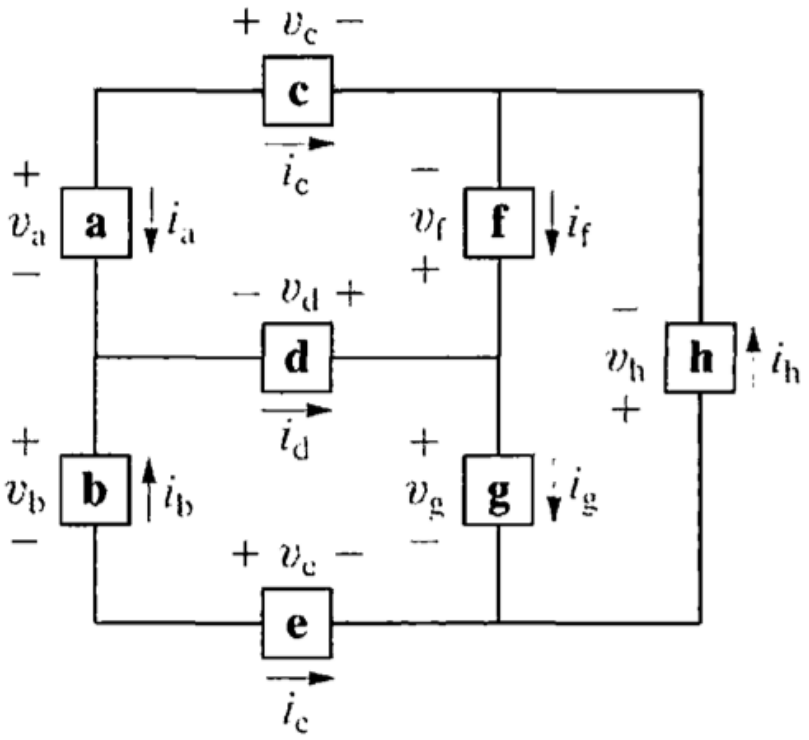
- The law of conservation of energy

$$\sum p = 0$$

- Energy is the capacity to do work, measured in joules (J).
- Mathematically,

$$w = \int_{t_0}^{t} p dt = \int_{t_0}^{t} vi dt$$

Example #5



Component	v(V)	i(A)
a	120	−10
b	120	9
c	10	10
d	10	1
e	−10	−9
f	−100	5
g	120	4
h	−220	−5

$$p_a = v_a i_a = (120)(-10) = -1200 \text{ W}$$

$$p_b = -v_b i_b = -(120)(9) = -1080 \text{ W}$$

$$p_c = v_c i_c = (10)(10) = 100 \text{ W}$$

$$p_d = -v_d i_d = -(10)(1) = -10 \text{ W}$$

$$p_e = v_e i_e = (-10)(-9) = 90 \text{ W}$$

$$p_f = -v_f i_f = -(-100)(5) = 500 \text{ W}$$

$$p_g = v_g i_g = (120)(4) = 480 \text{ W}$$

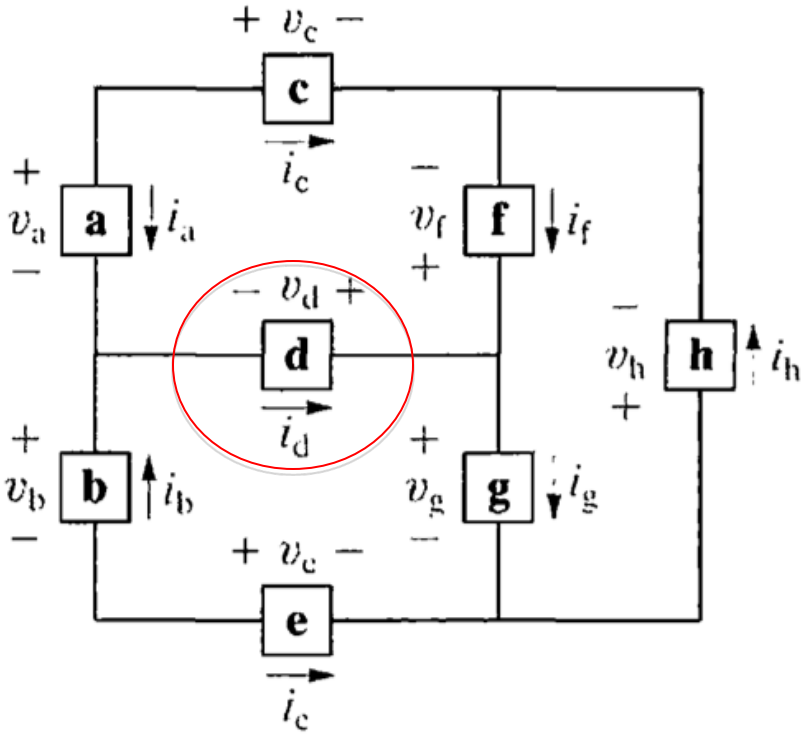
$$p_h = v_h i_h = (-220)(-5) = 1100 \text{ W}$$

$$p_{\text{supplied}} = p_a + p_b + p_d = -1200 - 1080 - 10 = -2290\ \text{W}$$

$$p_{\text{absorbed}} = p_c + p_e + p_f + p_g + p_h$$
$$= 100 + 90 + 500 + 480 + 1100 = 2270\ \text{W}$$

$$p_{\text{supplied}} + p_{\text{absorbed}} = -2290 + 2270 = -20\ \text{W}$$

Something is wrong—if the values for voltage and current in this circuit are correct, the total power should be zero!



Component	v(V)	i(A)
a	120	−10
b	120	9
c	10	10
d	10	1
e	−10	−9
f	−100	5
g	120	4
h	−220	−5

Summary

- Voltage
- Current
- Ideal basic circuit element
- Passive sign convention
- Power